

6ai	the spreading of the waves when it passes through a gap or around an obstacle	B1
6aii	wavelength of the electron must be comparable/of the same order of magnitude as the atomic spacing of the graphite	B1
6bi	Wavelength associated with a particle that is moving	B1
6bii	work done by p.d = increase in KE of electron $Vq = \frac{p^2}{2m}$ $p = \sqrt{2mVq}$ $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mVq}} = \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(120)(1.6 \times 10^{-19})}}$ $\lambda = 1.12 \times 10^{-10} \text{ m}$	C1 C1 C1 A1
6biii	When the p.d V is increased slightly, the <u>momentum of the electron would increase</u>. Hence, the de Broglie <u>wavelength of the electron would decrease</u>. Using $d \sin\theta = n\lambda$, the <u>angular separation between successive fringes would decrease</u>, and therefore <u>diameter of the concentric circles decreases</u>.	B1 C1 A1